

Teorema de la función implícita

Teorema 1 (de la función implícita). Sea $F: \Omega \subset \mathbb{R}^{n+m} \longrightarrow \mathbb{R}^n$ de clase $\mathcal{C}^1$ y $(a, b) \in \Omega$ ($a \in \mathbb{R}^n, b \in \mathbb{R}^m$) tal que $F(a, b) = 0 \wedge \det \left(\frac{\partial F_i}{\partial x_j}(a, b) \right)_{i,j=1}^n \neq 0$, entonces

- (1) $\exists U \subset \mathbb{R}^n$ abierto con $a \in U$.
- (2) $\exists V \subset \mathbb{R}^m$ abierto con $b \in V$.
- (3) $\exists ! g: V \longrightarrow U$ $\mathcal{C}^1$ tal que $\forall y \in V : F(g(y), y) = 0 \wedge g(b) = a$.

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